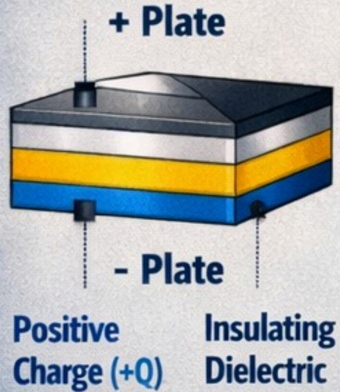
CAPACITORS IN PHYSICS

WHAT IS A CAPACITOR?

A capacitor is an electronic component that stores electrical energy by accumulating charge on two conductive plates separated by an inylating dielectric.

CAPACITOR STRUCTURE



Capacitance: $C = \frac{Q}{V}$

C = Capacitance (in Farads)

Q = Charge, (in Coulombs)

V = Voltage (in Volts)

STORED ENERGY

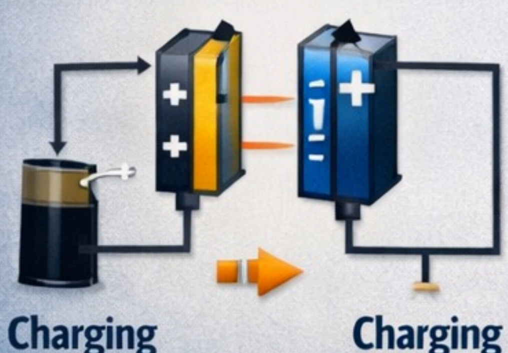
Energy: $E = \frac{1}{2} CV^2$

E = Energy Stored

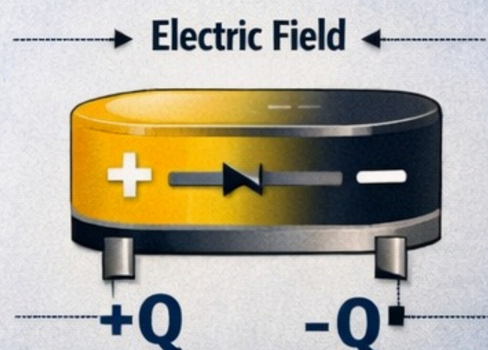
C = Capacitance

V = Voltage

CHARGING A CAPACITOR



STORED CHARGE



SUMMARY

104K = 1004F ±10%

+ 10 – 10 • 4–+10,000 (4 *Zeroa*)

× K = a10% Tolerance